

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M Tech (CSE) Sem -II University Examination, May-2011

EC – 712: RF & Wireless Communication Systems

Date: 20.05.11, Friday Time: 10:00 a.m to 01:00 p.m

Maximum Marks: 70

Instructions: (1) All questions are compulsory.

(2) Indicate clearly, the option you attempt along with its respective question number.

(3) Figures to right indicate marks

(4) Rough work is done in the last page of main answer book; don't write anything on the question paper.

Section I

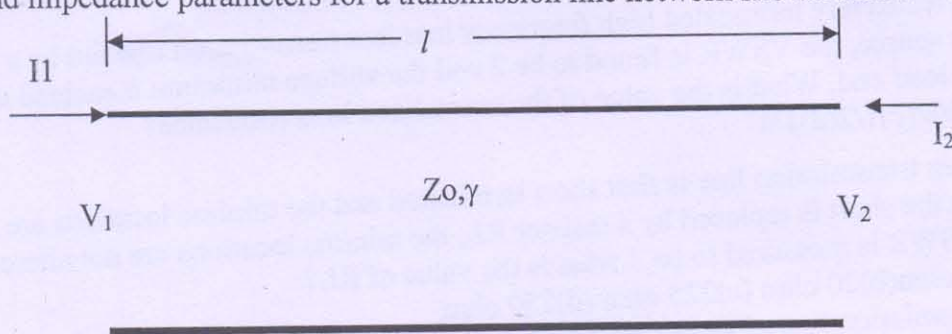
- Q-1**
- a Define the following terms : 5
Multi hop propagation, frequency selective fading, critical frequency, return loss, SWR
 - b On a resistively terminated high frequency loss less transmission line fed by a 500 MHz source, the VSWR is found to be 2 and the voltage minimum measured at 30cm from load end. What is the value of the normalized load resistance? 1
(a)3(b)2(c)1/2(d)1/4
 - c 75 ohm transmission line is first short terminated and the minima locations are noted. When the short is replaced by a resistor RL, the minima locations are not altered and the VSWR is measured to be 3.what is the value of RL? 1
(a)25 ohm(b)20 ohm (c)225 ohm (d)250 ohm
 - d A transmission line of length $n\lambda/4$ (n is integer) is short circuited at far end , the input impedance is 1
(a) 0 (b) infinite (c) Z_0 (d) $Z_0/2$
 - e A lossless transmission line having $Z_0=50 \Omega$, is terminated by load impedance of 3
(100-j150) Ω , using smith chart find
(1) line impedance at $d=0.3\lambda$ from load
(2) distance of first V_{min} from load
(3) SWR
- Q-2**
- a Discuss the layers of atmosphere 6
 - b Enlist various types of transmission line resonant circuit for $n=1$. Explain and compare short circuit and open circuit line resonator for their various parameter. 6
- OR**
- Q-2**
- a Design a low-pass filter for fabrication using micro strip lines on FR4 ($\epsilon_r=4.5$ 10
 $h=1.5\text{mm}$). The specification: cutoff frequency of 4GHz , third order, impedance 50 Ω , and a 3 dB equal-ripple characteristic (Using Richard's transform & Kuroda identity)
 - b An air filled rectangular cavity is made from piece of copper WR90 (0.9 X 0.4 in) 2
waveguide .If it resonates 9.379 GHz in TE₁₀₁ mode, find the required length c and the surface resistance of this resonator.(conductivity of copper $5.813 \times 10^7 \text{S/m}$)
- Q-3**
- a Based on distribution theory of transmission line, draw equivalent circuit of two wire 6
transmission line and derive wave equation for voltage & current for a lossless transmission line.
 - b Derive the image impedance using image parameter method 6
- OR**
- Q-3**
- a Design a low pass constant k -T section that has nominal impedance of 75 Ω and a 4
cut off frequency of 2 MHz. Plot its frequency response in frequency band of 100KHz

to 10 MHz

- b Design an air filled circular cylindrical cavity that resonate at 5GHz in TE₀₁₁ mode. It should be made up of copper and its height should be equal to its diameter. Find Q of this cavity. (conductivity of copper $5.813 \times 10^7 \text{ S/m}$) 4
- c Determine the element values of a resonant circuit to filter all the sinusoidal signals from 9 MHz to 11 MHz. This filter is to be connected between voltage source with negligible internal impedance and a communication system that has input impedance of 50 ohm 4

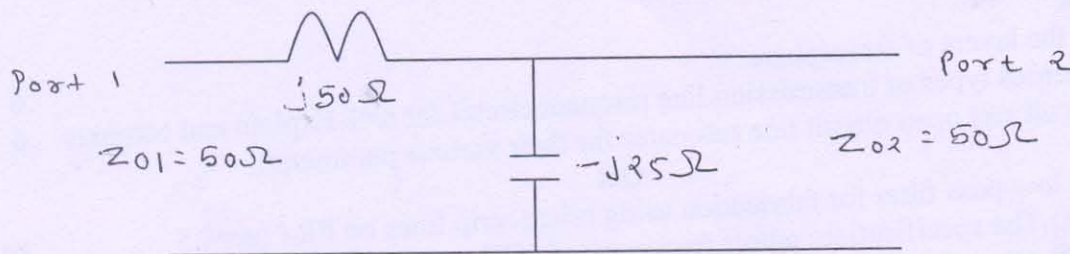
Section II

- Q-4 a Prove that for a hexagonal geometry, co-channel reuse ratio is given by $Q = (3N)^{0.5}$ 2
- b Explain frequency reuse concept 3
- c Define: Erlang, Peak hour, Cell dragging 3
- d Compare soft and hard handoff 2
- e How we can find the nearest co-channel cell 1
- Q-5 a Compare protocol model of ss7 with open system interconnection protocol model 6
- b Find impedance parameters for a transmission line network shown below 6



OR

- Q-5 a Explain channel assignment strategies. 6
- b Determine scattering parameters of the network shown below 6



- Q-6 a Explain Advanced Intelligent Network System architecture 6
- b Compare channel sharing and channel borrowing 6
- OR
- Q-6 a Compare second generation and third generation system 4
- b What is intelligent Cell? 2
- c Explain the different types of propagations 6